Core Banking Platform

Single Design Document

High Availability Architecture

Document ID:	SDD-CBP-2026-001
Version:	2.3
Date:	March 15, 2026
Classification:	Internal - Confidential
Owner:	Infrastructure Architecture Team
Status:	Approved

Version History

Version	Date	Author(s)	Description of Changes
2.3	03/15/2026	J. Martinez, A. Singh	Updated capacity numbers, Q1 2026 DR test results
2.2	11/20/2025	A. Singh	Db2 performance tuning, DR test documentation
2.1	08/10/2025	J. Martinez	Redis cache cluster addition
2.0	05/01/2025	R. Thompson	Major revision - GDPS migration complete
1.0	01/15/2024	R. Thompson	Initial document creation

Table of Contents

1 Purpose of this Document
- 1.1 Customer Background
- 1.2 Problem Statement

2 High Level Architecture
- 2.1 Architecture Overview
- 2.2 Design Principles
- 2.3 System Criticality Assessment

3 Infrastructure Components
- 3.1 Mainframe Infrastructure
- 3.2 Distributed Systems
- 3.3 Storage & Networking

4 High Availability Design
- 4.1 HA Architecture Pattern
- 4.2 Redundancy by Tier
- 4.3 Replication Architecture

5 Failover & Disaster Recovery
- 5.1 Automated Failover Mechanisms
- 5.2 Site Failover Procedure
- 5.3 Recovery Testing & Validation

6 Network Architecture
- 6.1 Security Zone Design
- 6.2 Cross-DC Connectivity

7 Data Flow & Processing
- 7.1 Transaction Processing Flow
- 7.2 Batch Processing Windows
- 7.3 Data Replication Topology

8 Monitoring & Observability
- 8.1 Monitoring Stack
- 8.2 Health Check Configuration
- 8.3 Alert Thresholds & Escalation

9 Security Controls
- 9.1 Encryption Standards
- 9.2 Access Control Framework
- 9.3 Compliance & Audit

10 Capacity Planning
- 10.1 Current Utilization
- 10.2 Growth Projections

11 Known Risks & Deviations

12 References

1 Purpose of this Document

The purpose of this document is to provide a comprehensive reference for the high availability architecture of the Example Organization's Core Banking Platform (CBP). It captures the design decisions, infrastructure topology, failover mechanisms, and operational procedures that ensure continuous service delivery for the bank's most critical transaction processing system.

This document is intended for architecture review boards, infrastructure operations teams, disaster recovery planners, audit and compliance functions, and any stakeholder requiring a detailed understanding of how CBP maintains availability. It serves as the authoritative reference during architecture reviews, DR planning exercises, and regulatory examinations.

1.1 Customer Background

Example Organization is a wholly-owned subsidiary of Example Holdings and a member of the Example Group, one of the largest banking groups in the world. Example Organization provides a broad range of financial products and services to individuals, small and mid-sized businesses, and corporate clients through its network of branches primarily in the northeastern United States.

The Core Banking Platform is the single most critical technology system within Example Organization's infrastructure. It processes approximately 45 million financial transactions daily across all banking channels including retail branches, ATMs, online banking, and mobile applications. Any disruption to this system directly impacts the bank's ability to serve its 12 million customers and generates regulatory scrutiny from the Office of the Comptroller of the Currency (OCC).

1.2 Problem Statement

Given the criticality of the Core Banking Platform, maintaining high availability is not merely a technical objective but a regulatory mandate. The OCC's Heightened Standards require that nationally chartered banks maintain robust business continuity capabilities for critical systems. Unplanned downtime carries a direct financial impact of $2.3 million per hour in lost transaction fees, customer compensation, and potential regulatory penalties.

This SDD documents the architecture that achieves 99.95% availability (less than 4.38 hours of unplanned downtime per year) through a combination of intra-site redundancy, automated failover mechanisms, and cross-site disaster recovery capabilities. The architecture must balance several competing concerns: zero data loss within the primary site, sub-30-second recovery point for cross-site failover, minimal transaction latency impact from redundancy mechanisms, and compliance with PCI-DSS, SOX, and OCC requirements.

2 High Level Architecture

2.1 Architecture Overview

The Core Banking Platform employs an Active-Passive architecture with hot standby, deployed across two geographically separated data centers. The primary site (DC-East, Newark, NJ) handles 100% of production traffic during normal operations, while the disaster recovery site (DC-West, Dallas, TX) maintains continuously replicated data and pre-warmed application instances ready for activation.

Within the primary site, every component is designed with local redundancy to eliminate single points of failure. The mainframe tier uses IBM Parallel Sysplex for active-active transaction processing. The distributed application tier runs N+2 redundancy, meaning the system can lose two of its eight application server instances while still operating at full capacity. The database tier uses GDPS (Geographically Dispersed Parallel Sysplex) for both local synchronous mirroring and cross-site asynchronous replication.

The decision to use Active-Passive rather than Active-Active across data centers was driven by data consistency requirements. Financial transaction processing demands strong consistency guarantees that are difficult to achieve with multi-site active writes given the 35ms round-trip latency between the two data centers. The asynchronous replication approach accepts a maximum 30-second recovery point objective in exchange for maintaining sub-100ms transaction latency for the majority of operations.

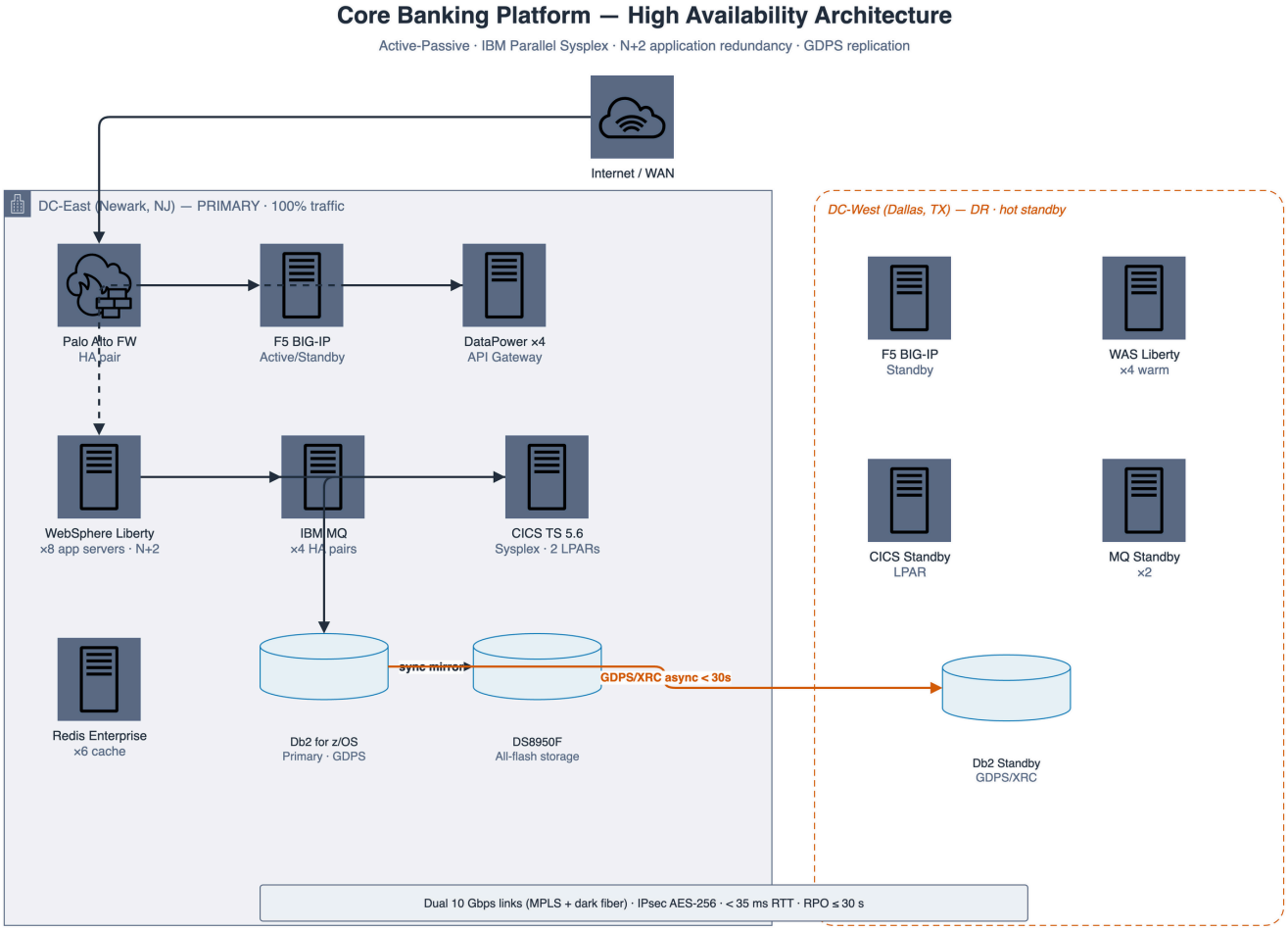


Figure 1: Core Banking Platform - High Availability Architecture Overview

2.2 Design Principles

The high availability architecture is built upon several foundational design principles that guide all infrastructure decisions:

No Single Point of Failure: Every component in the request path from customer transaction initiation through database commit must have at least one redundant counterpart. This applies to compute, network, storage, and power infrastructure. The architecture tolerates the simultaneous failure of any single component without service interruption.

Defense in Depth: Multiple independent layers of protection ensure that a failure in one protection mechanism does not leave the system vulnerable. For example, database availability is protected by local mirroring (PPRC), cross-site replication (XRC), periodic backups, and continuous transaction log archiving -- each operating independently.

Automated Recovery: Wherever possible, failover mechanisms operate automatically without human intervention. Manual steps are reserved only for cross-site failover decisions where the business impact of a false positive (unnecessary failover) could exceed the impact of a brief delay in recovery. All automated mechanisms are tested monthly.

Graceful Degradation: When a component fails, the system should degrade gracefully rather than failing completely. For example, if the Redis cache cluster becomes unavailable, the application servers automatically route balance queries directly to the database with slightly higher latency rather than returning errors to customers.

2.3 System Criticality Assessment

The Core Banking Platform has been assessed as a Tier-1 Business Critical system under the organization's internal technology risk framework. This classification drives the following recovery objectives and operational requirements:

Attribute	Requirement	Current Achievement
Availability SLA	99.95% (annual)	99.97% (trailing 12 months)
Recovery Time Objective (site)	< 15 minutes	12 minutes (last DR test)
Recovery Point Objective (site)	< 60 seconds	< 30 seconds (measured)
Recovery Time Objective (intra-site)	< 2 minutes	< 45 seconds (measured)
Recovery Point Objective (intra-site)	Zero data loss	Achieved (sync replication)
DR Test Frequency	Quarterly	Quarterly (last: Jan 22, 2026)
Maximum Transaction Latency P99	< 500ms	220ms (current P99)

The financial impact of an unplanned outage has been quantified at $2.3 million per hour, encompassing lost interchange fees, customer compensation, emergency staffing costs, and potential regulatory fines. Extended outages exceeding 4 hours trigger mandatory notification to the OCC and may result in consent order requirements for remediation.

3 Infrastructure Components

The Core Banking Platform infrastructure spans both mainframe and distributed computing environments. The transaction processing core runs on IBM z/OS, leveraging decades of investment in mainframe reliability and transaction integrity. Surrounding the mainframe core, a distributed tier of application servers, API gateways, and caching infrastructure handles channel connectivity, request routing, and performance optimization.

3.1 Mainframe Infrastructure

The mainframe environment runs on an IBM z15 T02 (model 3907) configured with 12,000 MIPS of processing capacity distributed across 8 Logical Partitions (LPARs). The z15 platform was selected for its industry-leading reliability characteristics including redundant processors, memory, and I/O paths, combined with hardware-assisted workload management and cryptographic acceleration.

Two LPARs are dedicated to CICS Transaction Server 5.6, which serves as the transaction processing engine. These LPARs participate in a Parallel Sysplex configuration with a shared Coupling Facility, enabling workload balancing and transparent failover between regions. The Sysplex configuration means that if one CICS LPAR fails, the other immediately absorbs the full transaction load with no perceptible impact to end users.

Db2 for z/OS 13.1 provides the primary database, configured in a Data Sharing Group that allows both Db2 members to access the same data simultaneously through Coupling Facility lock structures. This architecture eliminates database-level single points of failure within the primary site while maintaining full ACID transaction guarantees.

IBM MQ 9.3 provides reliable messaging between the distributed application tier and the mainframe CICS regions. Four Queue Managers are deployed in two highly-available pairs, ensuring that no single Queue Manager failure interrupts message flow. Messages are persisted to disk (persistent delivery) ensuring zero message loss even during a Queue Manager restart.

3.2 Distributed Systems

The distributed tier consists of WebSphere Liberty application servers that serve as the integration layer between external banking channels and the mainframe transaction engine. Eight Liberty instances are deployed in an N+2 configuration, meaning the system requires only six instances to handle peak load; two additional instances provide headroom for maintenance and unexpected failures.

IBM DataPower Gateway appliances provide the API gateway function at the perimeter, handling TLS termination, request validation, JWT token verification, schema enforcement, and rate limiting. Four physical DataPower appliances are deployed for redundancy and horizontal scaling, any two of which can handle the full production load.

Redis Enterprise 7.2 provides a distributed in-memory caching layer that dramatically reduces database load for frequently-accessed data such as account balances and session state. The six-node cluster (three primary shards with three replicas) can sustain the loss of any single node without service impact. Cache invalidation is triggered by database write operations through a change-data-capture mechanism.

3.3 Storage & Networking

Primary storage is provided by IBM DS8950F all-flash arrays configured with synchronous mirroring (PPRC) between

two arrays within the same data center. This configuration ensures zero data loss for any single storage array failure. Each array provides 120TB of raw capacity and 2.5 million IOPS, far exceeding current workload requirements.

Network infrastructure uses a defense-in-depth approach with multiple security zones separated by Palo Alto PA-7080 next-generation firewalls in a high-availability active-passive configuration. Traffic between zones is subject to strict policy enforcement with full deep-packet inspection. The network backbone provides 40Gbps of aggregate switching capacity with redundant paths to eliminate any network-level single points of failure.

4 High Availability Design

4.1 HA Architecture Pattern

The Core Banking Platform implements a tiered high availability design where each architectural layer employs the redundancy pattern most appropriate to its characteristics. The mainframe tier uses active-active processing through Parallel Sysplex. The application tier uses N+2 redundancy with health-check-driven load balancing. The database tier uses synchronous replication for local HA and asynchronous replication for cross-site DR.

This layered approach ensures that the most common failure scenarios -- a single process crash, a hardware component failure, or an operating system issue -- are handled automatically at the local level with recovery times measured in seconds. Only a complete site loss requires the more complex (and slower) cross-site failover procedure.

4.2 Redundancy by Tier

The following table summarizes the redundancy approach for each infrastructure tier, including the failure domains each mechanism protects against and the expected recovery time:

Tier	Pattern	Capacity	Recovery Time
Load Balancer	Active-Passive (HA Pair)	2 F5 appliances	< 10 seconds
API Gateway	Active-Active Pool	4 DataPower (need 2)	Immediate
Application	N+2 Redundancy	8 WAS (need 6)	< 30 seconds
Messaging	Multi-Instance QM	4 QMs in 2 HA pairs	< 60 seconds
Transaction Engine	Sysplex (Active-Active)	2 CICS LPARs	< 45 seconds
Database	GDPS Active-Passive	Primary + standby	< 2 minutes
Cache	Redis Cluster (3+3)	6 nodes	< 15 seconds

A critical design consideration is that the failure of any single tier must not cascade to other tiers. Circuit breaker patterns are implemented at each integration point to prevent a failing downstream dependency from consuming connection pools or thread resources in the calling tier. Each tier maintains independent health monitoring and recovery mechanisms.

4.3 Replication Architecture

Data replication is the cornerstone of the disaster recovery strategy. The platform employs a multi-level replication architecture that balances consistency, latency, and geographic separation. Within the primary site, GDPS/PPRC provides synchronous mirroring with zero recovery point objective -- every write is confirmed on both the primary and mirror storage before the application receives a commit acknowledgment.

Cross-site replication uses GDPS/XRC (Extended Remote Copy) which operates asynchronously to avoid imposing cross-data-center latency on every transaction. The XRC mechanism maintains a consistency group that ensures all replicated data at the DR site represents a consistent point-in-time, even though individual data sets may replicate at slightly different speeds. Current measurements show a typical replication lag of 12-18 seconds with a maximum observed lag of 28 seconds during peak processing periods.

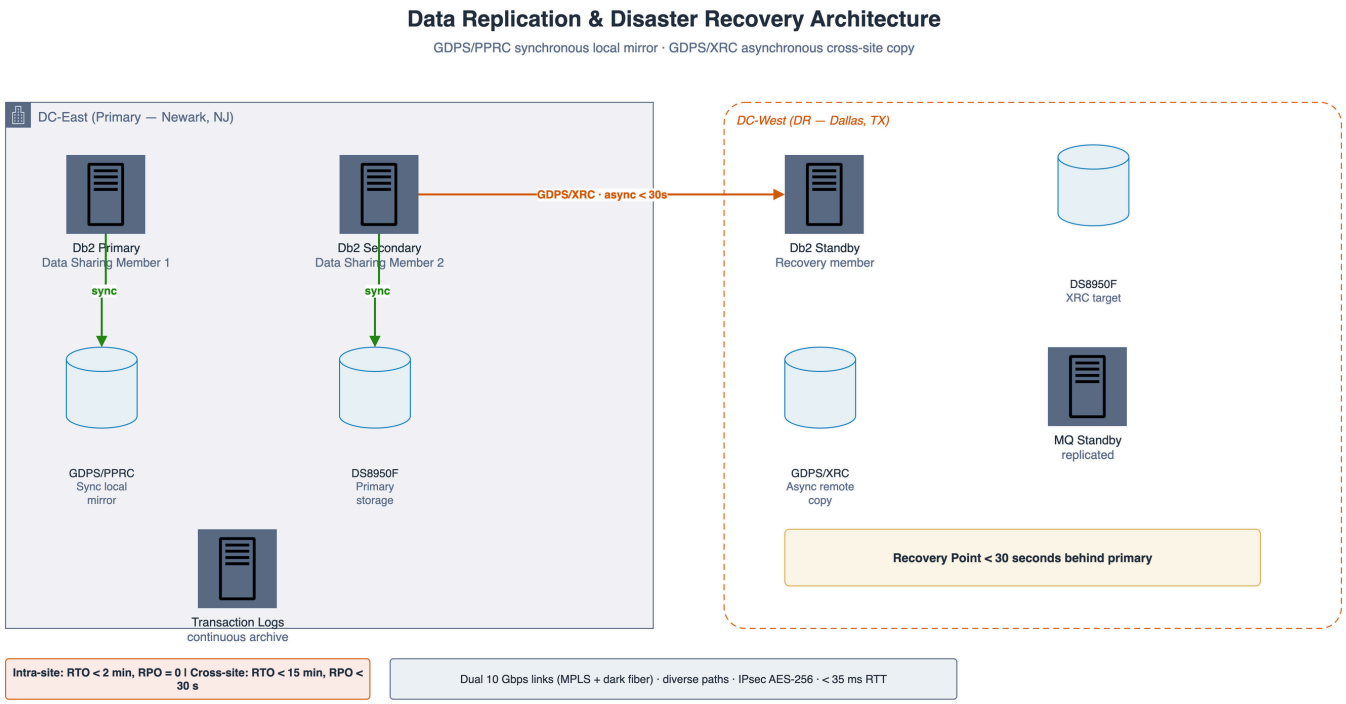


Figure 2: Data Replication and Disaster Recovery Architecture

5 Failover & Disaster Recovery

5.1 Automated Failover Mechanisms

The majority of component failures are handled by automated failover mechanisms that require no human intervention. These mechanisms operate at different layers of the stack and are tested monthly as part of the operational resilience program. The automation ensures that common failure scenarios -- such as a process crash, network interface failure, or hardware component failure -- are resolved within seconds.

At the load balancer tier, F5 BIG-IP uses a heartbeat mechanism between the active and standby units. If three consecutive heartbeats are missed (approximately 10 seconds), the standby unit assumes the active role and begins processing traffic. Connection mirroring ensures that established TCP connections are preserved across the failover, making the transition transparent to most clients.

At the application tier, the F5 load balancer monitors each WebSphere Liberty instance via HTTPS health checks to the /health/ready endpoint every 10 seconds. This endpoint performs a deep health check that validates database connectivity, MQ connectivity, and cache availability. If three consecutive checks fail, the instance is removed from the pool and traffic is redistributed to the remaining healthy instances. A new instance is simultaneously started by the orchestration layer.

At the mainframe tier, the z/OS Automatic Restart Manager (ARM) monitors CICS and Db2 address spaces. If an address space fails, ARM automatically restarts it, typically within 30-45 seconds. During this interval, the workload is handled by the surviving CICS LPAR through Sysplex Workload Manager routing.

5.2 Site Failover Procedure

A complete site failover from DC-East to DC-West is the most complex recovery scenario and is the only failover type that requires explicit human authorization. This decision point exists because an unnecessary site failover (false positive) carries significant operational risk and cost, including potential data loss of up to 30 seconds of transactions, a complex failback procedure, and disruption to batch processing schedules.

The site failover procedure is orchestrated by GDPS and follows a phased approach designed to ensure data consistency at each step. The total target recovery time is 15 minutes from the moment of declaration to the resumption of transaction processing at the DR site. Quarterly testing has consistently achieved this target, with the most recent test on January 22, 2026 completing in 12 minutes.

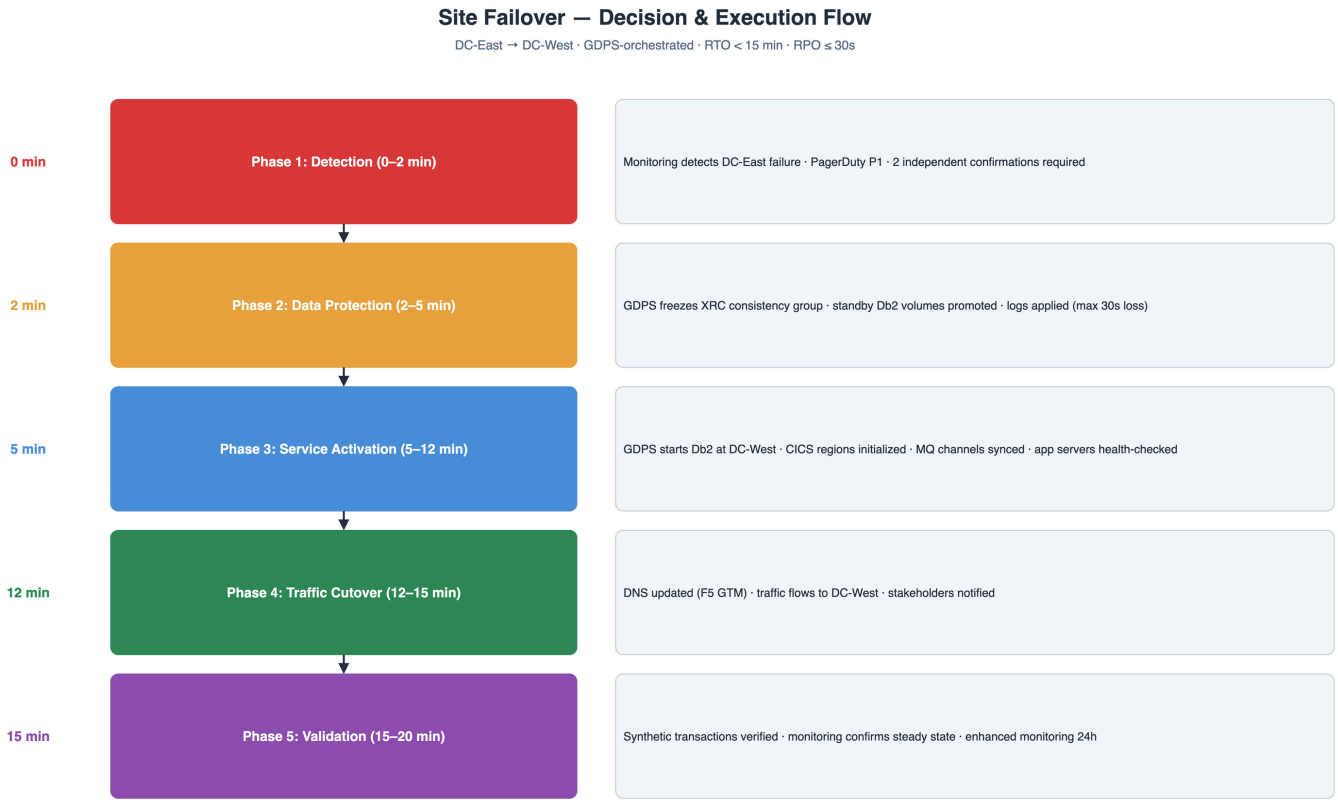


Figure 3: Site Failover Decision and Execution Flow

The detection phase relies on multiple independent monitoring systems to confirm a genuine site failure. A minimum of two independent confirmations are required before declaration: for example, loss of all Dynatrace agents at DC-East combined with loss of the cross-DC network heartbeat. This dual-confirmation requirement prevents false positives from triggering an unnecessary failover.

The data protection phase is the most critical step from a data integrity perspective. GDPS freezes the XRC consistency group to establish the latest consistent recovery point, then applies any pending transaction log records to the standby Db2 instance. The maximum data loss is bounded by the XRC replication lag at the moment of failure -- measured and monitored to remain below 30 seconds under all conditions.

5.3 Recovery Testing & Validation

The disaster recovery architecture is validated through a comprehensive testing program that includes quarterly full-site failover tests, monthly component failover tests, and weekly automated resilience checks. Each test is documented with pass/fail criteria aligned to the stated RTO and RPO objectives.

Test Type	Frequency	Last Execution	Result
Full site failover (DC-East to DC-West)	Quarterly	January 22, 2026	PASS (12 min RTO)
Individual component failover	Monthly	March 5, 2026	PASS (all within target)
Backup restore verification	Monthly	March 12, 2026	PASS
Automated failover (no human input)	Monthly	March 5, 2026	PASS

Tabletop exercise (scenario planning)	Semi-annual	December 10, 2025	3 action items

Test results are reviewed by the Architecture Review Board and any findings that could impact recovery capability are tracked to resolution. The July 2025 DR test revealed a CICS startup sequencing issue that caused the test to exceed the 15-minute RTO target (actual: 18 minutes). This was root-caused to an incorrect dependency ordering in the GDPS automation script and was corrected in August 2025. Subsequent tests have all completed within the 15-minute target.

6 Network Architecture

6.1 Security Zone Design

The network architecture implements a defense-in-depth model with multiple security zones, each enforcing specific access controls and traffic inspection policies. Traffic traversal between zones is controlled by Palo Alto next-generation firewalls that perform deep packet inspection, application-layer filtering, and threat prevention. The zone architecture ensures that a compromise in one zone cannot easily propagate to adjacent zones.

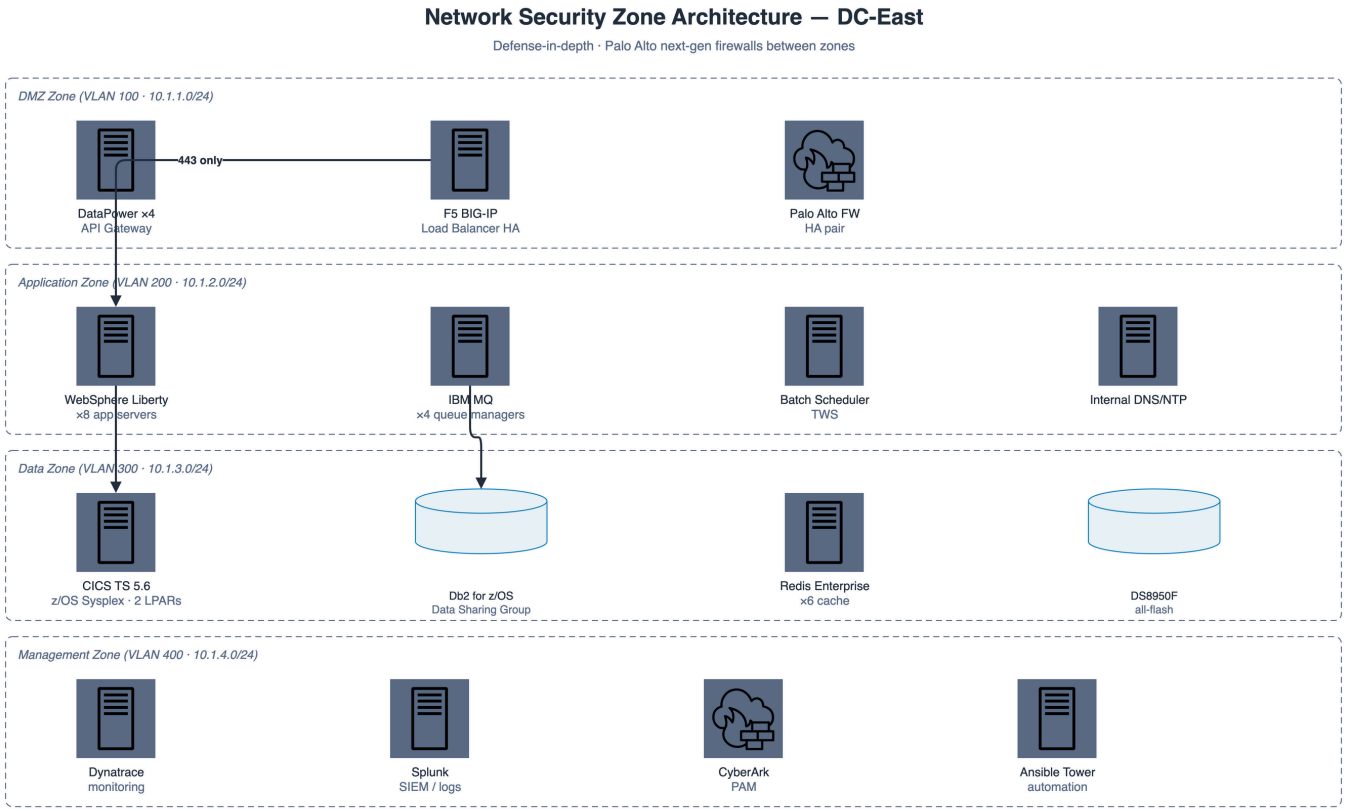


Figure 4: Network Security Zone Architecture - DC-East

The DMZ zone (VLAN 100, 10.1.1.0/24) hosts the externally-facing API gateway appliances and the load balancer virtual servers. Only HTTPS (port 443) traffic is permitted inbound from the internet, and all traffic is subject to Web Application Firewall (WAF) inspection before reaching the API gateways. The DMZ has no direct path to the data zone; all communication must traverse the application zone.

The Application zone (VLAN 200, 10.1.2.0/24) contains the WebSphere Liberty servers, IBM MQ queue managers, and batch scheduling infrastructure. This zone can communicate with the DMZ zone (to receive requests from the API gateway) and with the Data zone (to access databases and mainframe services). No direct internet connectivity is available in this zone.

The Data zone (VLAN 300, 10.1.3.0/24) is the most restricted network segment, containing the CICS mainframe LPARs, Db2 databases, Redis cache cluster, and storage arrays. Access to this zone is permitted only from the Application zone on specific ports corresponding to database protocols (Db2 port 446), cache protocols (Redis port 6379), and message

queue protocols. Administrative access requires traversal through the Management zone via a privileged access management (PAM) gateway.

6.2 Cross-DC Connectivity

Connectivity between DC-East and DC-West is provided by two diverse 10Gbps links from different telecommunications providers (AT&T MPLS and Zayo dark fiber). The links follow geographically separate physical paths with no shared conduit, building entry, or central office, ensuring that a single physical event (construction dig-up, building fire) cannot sever both connections simultaneously.

Both links carry production replication traffic encrypted with IPsec using AES-256-GCM and IKEv2 key exchange. Bidirectional Forwarding Detection (BFD) monitors link health at 300ms intervals, enabling automatic failover between links within approximately 1 second of detecting a link failure. The measured round-trip latency is 32ms on the primary MPLS link and 34ms on the secondary dark fiber link.

Under normal conditions, both links are active with GDPS/XRC replication traffic on the primary link and management/monitoring traffic on the secondary. In a link failure scenario, all traffic converges onto the surviving link, which has sufficient bandwidth to handle the combined load (current peak utilization is 4.2Gbps across both links).

7 Data Flow & Processing

7.1 Transaction Processing Flow

A typical financial transaction flows through seven distinct processing stages from customer initiation to database commit. The end-to-end latency budget is 500ms at P99, though the actual measured P99 latency is significantly better at 220ms. Each stage adds value through validation, enrichment, or transformation, and each has independent timeout and retry characteristics.

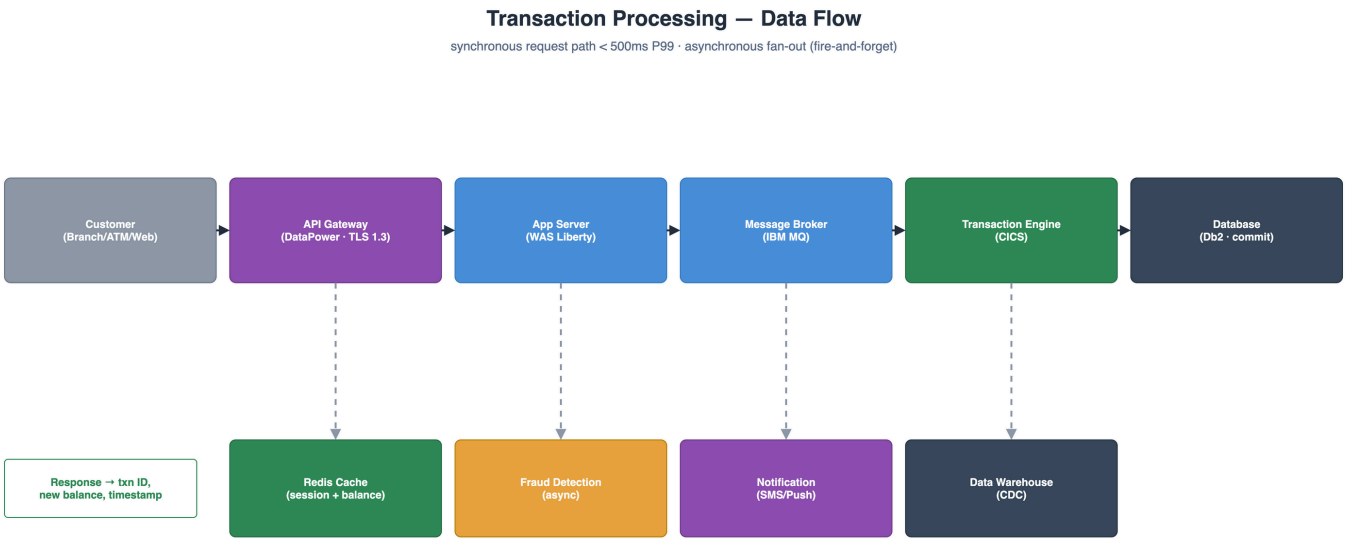


Figure 5: Transaction Processing Data Flow

The flow begins when a customer initiates a transaction through any banking channel (branch teller system, ATM, online banking, or mobile application). The channel system formats the request and sends it to the API Gateway (DataPower) over TLS 1.3. DataPower validates the request schema, verifies the JWT authentication token, applies rate limiting rules, and routes the request to an available WebSphere Liberty application server.

The application server performs business-level validation including customer authentication verification, transaction limit checks, and account status validation. It queries the Redis cache for the current account balance (cache hit rate: 94%) and initiates an asynchronous fraud scoring request via MQ. If the fraud score exceeds the configured threshold, the transaction is held for manual review rather than declined automatically, preserving the human-in-the-loop principle for high-impact decisions.

Once validated, the transaction is submitted to CICS via IBM MQ using persistent, transacted delivery. CICS executes the core banking logic: debiting the source account, crediting the destination account, posting to the general ledger, and creating an immutable audit record. The entire CICS operation executes within a single unit of work (UOW), ensuring atomicity. Upon successful commit in Db2, the response propagates back through the chain to the customer.

7.2 Batch Processing Windows

In addition to real-time transaction processing, the Core Banking Platform executes a series of batch processing jobs during off-peak hours. These jobs handle operations that are computationally intensive or require processing across the full account population, such as interest calculations, statement generation, and regulatory reporting extracts.

The batch processing window begins at 22:00 ET with End-of-Day (EOD) processing and must complete all jobs by 06:00 ET when business-day transaction volumes resume. The EOD process itself typically completes in 2.5 hours, followed by interest calculation (1.5 hours), statement generation (2 hours), and regulatory extracts (45 minutes). Careful scheduling ensures that dependent jobs execute in the correct sequence while maximizing parallelism for independent workstreams.

Batch processing is a critical consideration for disaster recovery planning because an unscheduled failover during the batch window may require restarting jobs from their last checkpoint. All batch jobs are designed with restart/recovery capabilities that allow them to resume from the most recent commit point rather than reprocessing from the beginning.

7.3 Data Replication Topology

Data flows out of the Core Banking Platform through multiple replication pathways to support downstream consumers. The primary replication mechanism is IBM Change Data Capture (CDC), which reads the Db2 transaction log and publishes change events to downstream systems including the Data Warehouse, Customer Data Platform, and regulatory reporting systems.

The CDC mechanism operates asynchronously and does not impact transaction processing latency. Under normal conditions, the CDC lag is less than 5 minutes, meaning downstream systems see transaction data within 5 minutes of the commit in the primary database. This latency is acceptable for all current downstream use cases, which are either analytical (tolerating minutes of delay) or batch-oriented (processing on a daily schedule).

8 Monitoring & Observability

8.1 Monitoring Stack

The observability architecture combines multiple specialized monitoring tools to provide comprehensive visibility across the mainframe and distributed tiers. Dynatrace serves as the primary Application Performance Monitoring (APM) platform for the distributed tier, providing full-stack monitoring from infrastructure metrics through application-level transaction tracing. IBM OMEGAMON provides equivalent visibility into the mainframe environment including CICS region health, Db2 performance, and MQ throughput.

All monitoring data flows into a centralized Splunk instance for log correlation, security event analysis, and long-term retention. PagerDuty serves as the alerting and incident management platform, receiving alerts from both Dynatrace and OMEGAMON and routing them to the appropriate on-call personnel based on severity and affected component.

8.2 Health Check Configuration

Every component in the platform is subject to automated health monitoring at intervals appropriate to its criticality and expected failure characteristics. The health check architecture distinguishes between shallow checks (is the process running?) and deep checks (can the component successfully process a request end-to-end?). Deep checks are preferred wherever the additional overhead is acceptable.

Component	Check Type	Interval	Unhealthy After
API Gateway (DataPower)	TCP port 443 open	5 seconds	3 failures (15s)
App Server (WAS Liberty)	HTTP GET /health/ready	10 seconds	3 failures (30s)
CICS Region	CEMT INQUIRE TASK	30 seconds	2 failures (60s)
Db2 Member	SQL SELECT 1 FROM SYSIBM.DUAL	15 seconds	2 failures (30s)
MQ Queue Manager	MQSC DISPLAY QMSTATUS	30 seconds	2 failures (60s)
Redis Node	PING command	5 seconds	3 failures (15s)
F5 BIG-IP	SNMP + iControl REST	10 seconds	2 failures (20s)
Cross-DC Link	BFD + ICMP echo	300ms	3 failures (~1s)

8.3 Alert Thresholds & Escalation

Alert thresholds are calibrated based on historical baseline data and business impact analysis. Each alert has a clear severity classification (P1 through P4) with defined response times and escalation paths. Warning-level alerts (P3) indicate that a metric is approaching a concerning threshold and requires investigation during business hours. Critical alerts (P1) indicate an active or imminent service impact requiring immediate response regardless of time of day.

The on-call rotation for the Core Banking Platform is staffed 24/7 with a primary engineer and a secondary backup. P1 alerts require a 5-minute response time, with escalation to the team lead at 30 minutes and to the VP of Infrastructure at 2 hours if unresolved. All P1 incidents trigger a post-incident review within 5 business days, with findings tracked to resolution.

9 Security Controls

9.1 Encryption Standards

All data handled by the Core Banking Platform is encrypted both in transit and at rest, using cryptographic algorithms and key lengths that meet or exceed current NIST recommendations and PCI-DSS requirements. Encryption keys are managed through a combination of IBM Integrated Cryptographic Service Facility (ICSF) for mainframe workloads and Thales Hardware Security Modules (HSMs) for distributed systems.

Data in transit between external clients and the API gateway uses TLS 1.3 exclusively, with TLS 1.2 as a minimum for internal service-to-service communication. Cipher suite selection prioritizes forward secrecy (ECDHE key exchange) and authenticated encryption (AES-256-GCM). Cross-data-center replication traffic is encrypted using IPsec with AES-256-GCM and Perfect Forward Secrecy via IKEv2.

Data at rest is encrypted at the storage layer using AES-256. On the mainframe, Db2 native encryption protects database tablespaces with keys stored in the ICSF CKDS (Cryptographic Key Data Set). On the distributed tier, Linux volumes use LUKS with keys managed by HashiCorp Vault. Encryption key rotation follows an annual cycle for data encryption keys and a bi-annual cycle for key encryption keys.

9.2 Access Control Framework

Access to the Core Banking Platform infrastructure follows the principle of least privilege with multi-layered authentication and authorization controls. No individual has standing privileged access to production systems; all administrative access is mediated through CyberArk Privileged Access Management (PAM) which provides just-in-time credential checkout, session recording, and automatic credential rotation.

On the mainframe, RACF (Resource Access Control Facility) provides the security layer for z/OS resources, CICS regions, and Db2 objects. All RACF rules follow a default-deny model where access must be explicitly granted. On the distributed tier, Active Directory group membership controls operating system access, with service accounts managed separately and rotated every 90 days automatically.

9.3 Compliance & Audit

The Core Banking Platform is subject to multiple compliance frameworks that impose specific requirements on infrastructure security, data protection, and operational procedures. PCI-DSS Level 1 certification is maintained through annual assessments by a Qualified Security Assessor (QSA), covering all components that process, store, or transmit cardholder data. SOX Section 404 compliance requires documented change control procedures and segregation of duties for systems supporting financial reporting.

OCC Heightened Standards impose additional requirements specific to nationally chartered banks, including mandatory disaster recovery testing, documented risk assessments, and explicit board-level oversight of technology risk. The FFIEC Cybersecurity Assessment Tool (CAT) is used to evaluate the bank's cybersecurity preparedness, with the Core Banking Platform currently rated at the 'Advanced' maturity level across all assessment domains.

10 Capacity Planning

10.1 Current Utilization

Capacity planning for the Core Banking Platform follows a proactive model where resource utilization is continuously monitored against defined thresholds. The target operating range is 50-70% of peak capacity under normal conditions, providing a 30-50% buffer for unexpected demand spikes, degraded-mode operation (fewer healthy instances), and planned growth before the next capacity augmentation.

Resource	Total Capacity	Average Use	Peak Use
Mainframe MIPS	12,000	5,880 (49%)	8,400 (70%)
App Server CPU (cores)	64	25 (39%)	38 (59%)
App Server Memory	512 GB	205 GB (40%)	320 GB (63%)
Database Storage	200 TB	130 TB (65%)	145 TB (73%)
Network Cross-DC	20 Gbps (2x10G)	2.1 Gbps (11%)	4.2 Gbps (21%)
Redis Cache Memory	96 GB	42 GB (44%)	58 GB (60%)

10.2 Growth Projections

Transaction volume is growing at approximately 15% year-over-year driven by increased digital banking adoption and new product launches. Database storage consumption is growing faster at 25% year-over-year due to expanded regulatory retention requirements and the increasing richness of transaction metadata. Based on these growth rates, the next significant capacity investment will be required in Q3 2026 for database storage, and Q1 2027 for mainframe MIPS.

The capacity planning cycle runs quarterly, producing a rolling 18-month forecast that accounts for organic growth, planned product launches, regulatory changes, and technology refresh cycles. Capital expenditure requests for infrastructure augmentation are submitted 6 months in advance of projected threshold breach to account for procurement and installation lead times, particularly for mainframe hardware.

11 Known Risks & Deviations

The following risks and deviations from the organization's standard HA reference patterns have been identified, assessed, and formally accepted through the Architecture Review Board (ARB). Each item includes a mitigation strategy and an owner responsible for tracking resolution where applicable.

R-001: Redis cache cluster is not replicated cross-DC: The Redis cache layer exists only at DC-East with no equivalent at DC-West. In a site failover scenario, all cache data is lost and must be rebuilt from the database. The impact is elevated database load for approximately 15-20 minutes during cache warm-up. This has been tested during DR exercises and confirmed not to exceed database capacity. Status: Accepted (ARB-2025-055).

R-002: Manual DNS failover step in site recovery: The cross-site failover procedure includes a manual DNS update step that adds approximately 45 seconds to the overall recovery time. An automated solution using F5 Global Traffic Manager (GTM) is currently in implementation and expected to eliminate this manual step by Q3 2026. Status: In Progress.

R-003: Cross-site RPO is not zero: Due to the physical distance between data centers (1,400 miles) and the resulting network latency (35ms RTT), synchronous replication across sites is not feasible without unacceptable transaction latency impact. The business has formally accepted the < 30 second RPO for cross-site failover scenarios. Status: Accepted (ARB-2024-078).

R-004: Mainframe skills shortage risk: The pool of engineers with deep z/OS, CICS, and Db2 expertise is shrinking as experienced staff approach retirement. A cross-training program is underway to develop the next generation of mainframe specialists, with 4 engineers currently in an 18-month development program. Status: In Progress.

12 Operational Procedures

12.1 Change Management

All changes to the Core Banking Platform follow the enterprise change management process governed by ITIL v4 principles. Changes are classified into three categories: standard changes (pre-approved, low-risk modifications such as parameter tuning or log level adjustments), normal changes (requiring Change Advisory Board review and a scheduled implementation window), and emergency changes (expedited approval with mandatory post-implementation review).

The weekly Change Advisory Board (CAB) meets every Tuesday to review proposed changes for the following week. Each change request must include a detailed implementation plan, a tested rollback procedure, a risk assessment, and evidence of successful testing in the pre-production environment. Changes to Tier-1 systems like CBP receive enhanced scrutiny and require sign-off from both the technical lead and the system owner.

Change freeze periods are enforced 48 hours before month-end processing, quarter-end reporting deadlines, and regulatory filing dates. Extended freezes apply during the year-end processing period (December 28 through January 3) and during scheduled regulatory examinations. Emergency changes during freeze periods require VP-level approval and carry a mandatory post-implementation review regardless of outcome.

12.2 Incident Management

Incidents affecting the Core Banking Platform are managed through a structured process that emphasizes rapid detection, clear communication, and thorough root cause analysis. The severity classification determines response times, communication frequency, and escalation paths:

Severity	Definition	Response	Escalation
SEV-1	Complete service outage	5 minutes	Immediate bridge call
SEV-2	Significant degradation (>10%)	15 minutes	30-min to team lead
SEV-3	Minor component impact	1 hour	Next business day
SEV-4	No customer impact	Next business day	Weekly review

SEV-1 incidents trigger an immediate conference bridge with representation from all potentially affected technology teams. The Incident Commander role is assumed by the senior on-call engineer, who is responsible for coordinating the response, maintaining a timeline of actions, and providing regular updates to business stakeholders. Updates to the executive team are provided every 15 minutes during SEV-1 events.

All SEV-1 and SEV-2 incidents require a Post-Incident Review (PIR) within 5 business days of resolution. The PIR follows a blameless methodology focused on identifying systemic improvements rather than individual fault. Action items from PIRs are tracked in the risk register and reviewed at monthly operational review meetings.

12.3 Patching & Maintenance

The patching strategy balances the need for timely security updates against the risk of introducing instability into a Tier-1 production system. Each component type follows a defined patching cadence with specific maintenance windows:

Component	Cadence	Window	Method
z/OS System Software	Quarterly	Sunday 02:00-06:00 ET	Rolling LPAR
CICS Transaction Server	Quarterly	Sunday 02:00-06:00 ET	Rolling LPAR
Db2 for z/OS	Quarterly	Extended Sunday window	Maintenance mode
WebSphere Liberty	Monthly	Sunday 02:00-04:00 ET	Rolling (node by node)
Redis Enterprise	Monthly	Sunday 02:00-03:00 ET	Rolling restart
F5 BIG-IP	Quarterly	Sunday 02:00-06:00 ET	HA failover patching
Palo Alto Firewall	Quarterly	Sunday 02:00-06:00 ET	HA failover patching
Linux OS (distributed)	Monthly	Sunday 02:00-04:00 ET	Ansible rolling

Critical security patches (CVE score 9.0 or higher) that affect externally-exposed components are expedited outside the normal cadence. These follow the emergency change process with a target of 72 hours from vulnerability disclosure to production deployment. The security team maintains a vulnerability management dashboard that tracks patch status across all CBP components.

12.4 Capacity Management

Capacity reviews are conducted quarterly, producing a rolling 18-month forecast that informs infrastructure investment decisions. The review process analyzes historical utilization trends, correlates them with business growth metrics (transaction volume, customer count, product launches), and projects future resource requirements with confidence intervals.

Threshold-based alerting provides early warning when resource utilization approaches planning limits. Warning alerts fire at 70% sustained utilization (indicating the need to schedule a capacity augmentation), while critical alerts fire at 85% (indicating an urgent need that may require temporary measures such as traffic shedding or workload deferral while permanent capacity is provisioned).

13 Dependency Mapping

13.1 External Dependencies

The Core Banking Platform has critical dependencies on several external systems and networks. Each dependency has been assessed for its impact on CBP availability and appropriate mitigation strategies have been implemented. The following external systems represent the highest-risk dependencies due to their synchronous nature or their role in critical transaction paths.

SWIFT Network: International wire transfers depend on connectivity to the SWIFT network via FIN messaging. SWIFT access is provided through two independent SWIFT Alliance Lite2 connections from different network entry points. If both connections fail, international transfers queue locally and are transmitted when connectivity restores. Average queue time during the last SWIFT outage (September 2025) was 47 minutes.

FedWire and ACH Networks: Domestic wire transfers use FedWire via the Federal Reserve's FedLine Direct connection. ACH transactions are processed through the Federal Reserve's FedACH system and Nacha's Electronic Payments Network (EPN). Both connections are redundant with automatic failover. Processing windows are defined by the Federal Reserve schedule and are not controllable by the bank.

Card Networks (Visa, Mastercard, Amex): Card authorization requests are routed to the appropriate card network in real-time. Each network connection uses ISO 8583 messaging over dedicated VPN tunnels. Stand-In Processing (STIP) is configured to allow the bank to approve transactions locally when network connectivity is temporarily unavailable, using local risk parameters and pre-authorized limits.

13.2 Internal Dependencies

Within the organization's technology landscape, CBP has synchronous dependencies on several internal services. The most critical is the Customer Identity Service (CIS), which validates authentication tokens for every transaction request. CIS is deployed as a clustered service with its own HA architecture, and CBP implements a 5-minute token cache to survive brief CIS outages without impacting transaction processing.

The Fraud Detection Engine (FDE) receives real-time transaction data via asynchronous messaging and returns risk scores. Because this integration is asynchronous (fire-and-forget from CBP's perspective), FDE unavailability does not block transaction processing. However, during FDE outages, transactions proceed without real-time fraud scoring, which increases the bank's fraud exposure. A compensating control automatically reduces transaction limits by 50% when FDE scoring is unavailable for more than 5 minutes.

The Anti-Money Laundering (AML) system receives transaction data for screening against sanctions lists and suspicious activity patterns. This is a synchronous dependency for wire transfers exceeding $10,000 (regulatory requirement for Currency Transaction Reports). For these transactions, a 5-second timeout is configured; if AML screening cannot complete within this window, the transaction is queued for manual review rather than being automatically approved, ensuring regulatory compliance is maintained even during system degradation.

13.3 Dependency Failure Impact Analysis

Dependency	Failure Impact	Detection	Mitigation

CIS (Identity)	Auth failures	Health check (10s)	Token cache (5 min)
Fraud Detection	Unscored transactions	MQ depth monitor	Reduced limits
AML Screening	Wire transfer delays	Response timeout	Manual queue
SWIFT Network	Int'l transfer delays	Heartbeat monitor	Local queue
Card Networks	Auth decline spike	Error rate alarm	STIP processing
DNS (Akamai)	Resolution failure	Synthetic checks	Local cache + hosts
NTP (Internal)	Clock drift risk	Offset monitoring	Local clock (30min)

14 Backup & Data Protection

14.1 Backup Architecture

The backup strategy for the Core Banking Platform is designed around the principle that no single data loss event should result in unrecoverable data. Multiple independent backup mechanisms operate concurrently, each protecting against different failure scenarios. The strategy encompasses database backups, configuration backups, transaction log archiving, and full system image preservation.

Database backups use IBM's FlashCopy technology for full backups and online incremental backups for point-in-time recovery between full backup cycles. FlashCopy creates an instantaneous consistent copy of the production data without impacting online transaction processing. The copy is then transferred to a secondary storage system and subsequently to tape for long-term retention.

Data Type	Method	Frequency	Retention
Db2 Full Backup	FlashCopy	Daily at 02:00 ET	30 days disk, 7yr tape
Db2 Incremental	Online backup	Every 4 hours	7 days
Transaction Logs	Continuous archive	Real-time	90 days disk
MQ Queue Data	dmpmqcfg + media	Daily	30 days
WAS Configuration	wsadmin export	On change + daily	90 days
Redis Snapshots	RDB + AOF	Every 15 minutes	7 days
System Configuration	Ansible/Git	On every change	Unlimited (Git)
Network Configuration	Automated export	On change + daily	1 year

14.2 Recovery Procedures

Recovery procedures are documented in detailed runbooks and tested regularly as part of the DR testing program. Each procedure is designed to be executable by any qualified on-call engineer using step-by-step instructions, reducing dependency on specific individuals and enabling recovery during off-hours when senior staff may not be immediately available.

For the most common recovery scenario -- a single Db2 member failure -- recovery is fully automatic through the GDPS framework. The standby member is activated, connections are rerouted, and processing resumes within 2 minutes. No manual intervention or backup restoration is required because the synchronous PPRC mirror provides a current copy of all data.

For logical data corruption (such as an erroneous batch job that corrupts account balances), recovery requires point-in-time restoration from the most recent clean backup followed by selective replay of transaction logs up to the point immediately before the corruption occurred. This procedure has been tested and validated with a recovery time of approximately 45 minutes for a single tablespace and 4 hours for a full database restoration.

14.3 Backup Validation

Backups are only valuable if they can be successfully restored. The backup validation program performs regular restore tests to confirm backup integrity and measure actual recovery times. Monthly tests restore a randomly selected tablespace to an isolated test environment and validate data integrity through checksum comparison with the production database. Quarterly tests perform a full database restoration to validate end-to-end recovery capability.

All backup validation results are recorded in the operational metrics dashboard and any failures trigger immediate investigation and remediation. The current backup success rate is 99.97% (3 failures in the past 12 months, all due to tape media issues that were resolved by media replacement and re-backup).

15 Known Risks & Deviations

The following risks and deviations from the organization's standard HA reference patterns have been identified, assessed, and formally accepted through the Architecture Review Board (ARB). Each item includes a mitigation strategy and an owner responsible for tracking resolution where applicable.

R-001: Redis cache cluster is not replicated cross-DC: The Redis cache layer exists only at DC-East with no equivalent at DC-West. In a site failover scenario, all cache data is lost and must be rebuilt from the database. The impact is elevated database load for approximately 15-20 minutes during cache warm-up. This has been tested during DR exercises and confirmed not to exceed database capacity. Status: Accepted (ARB-2025-055).

R-002: Manual DNS failover step in site recovery: The cross-site failover procedure includes a manual DNS update step that adds approximately 45 seconds to the overall recovery time. An automated solution using F5 Global Traffic Manager (GTM) is currently in implementation and expected to eliminate this manual step by Q3 2026. Status: In Progress.

R-003: Cross-site RPO is not zero: Due to the physical distance between data centers (1,400 miles) and the resulting network latency (35ms RTT), synchronous replication across sites is not feasible without unacceptable transaction latency impact. The business has formally accepted the < 30 second RPO for cross-site failover scenarios. Status: Accepted (ARB-2024-078).

R-004: Mainframe skills shortage risk: The pool of engineers with deep z/OS, CICS, and Db2 expertise is shrinking as experienced staff approach retirement. A cross-training program is underway to develop the next generation of mainframe specialists, with 4 engineers currently in an 18-month development program. Status: In Progress.

R-005: DR site is single-hall design: DC-West currently operates from a single data hall within the facility, meaning that a localized event within that hall (fire suppression activation, cooling failure) could impact the entire DR capability. A second-hall expansion project is approved and scheduled for completion in Q4 2026. Until then, the risk is mitigated by enhanced fire detection and suppression systems and a pre-negotiated contract for emergency colocation space. Status: In Progress.

R-006: Single storage vendor dependency: Both the primary and DR sites use IBM DS8950F storage arrays, creating vendor concentration risk. If a firmware defect affects the DS8950F platform, both sites could be impacted simultaneously. Mitigation includes maintaining firmware versions one release apart between sites (never upgrading both concurrently) and periodic evaluation of multi-vendor strategies. Status: Accepted (ARB-2025-102).

R-007: MQ replication lag during peak processing: During month-end peak processing periods, IBM MQ replication between DC-East and DC-West can experience lag up to 45 seconds (exceeding the normal < 5 second threshold). This means that in a site failover during peak, some messages may be duplicated when replayed from the source queue. Application-level idempotency checks are the primary mitigation. A queue manager performance tuning exercise is planned for Q2 2026. Status: Planned.

16 Testing & Validation Strategy

16.1 Resilience Testing Program

The Core Banking Platform participates in a comprehensive resilience testing program designed to validate that all high availability mechanisms function correctly under realistic failure conditions. The program operates on a continuous cycle of planning, execution, analysis, and improvement, with tests scheduled at varying frequencies depending on their scope and potential impact.

Chaos engineering principles are applied through controlled fault injection exercises that simulate specific failure scenarios in the production environment during low-traffic periods. These exercises validate not only the technical failover mechanisms but also the human processes: monitoring alerts fire correctly, on-call personnel respond within SLA, and communication procedures are followed. Each exercise produces a detailed report documenting the timeline of events, any deviations from expected behavior, and improvement recommendations.

16.2 Performance Testing

Performance baselines are established through quarterly load testing exercises that validate system behavior under normal, elevated, and peak transaction volumes. The load testing environment is a scaled replica of production (50% capacity) with synthetic transaction generators that produce realistic workload patterns including the correct mix of transaction types, account access patterns, and temporal distribution.

The most recent performance test (February 2026) validated that the system maintains P99 latency below 500ms at 120% of current peak volume (3,600 TPS), providing confidence that the platform can absorb organic growth for the next 12-18 months without requiring capacity augmentation. Stress testing to 200% of peak (6,000 TPS) identified the application server CPU as the first resource to saturate, confirming the expected bottleneck and validating the capacity planning model.

16.3 Security Testing

Security testing encompasses both automated scanning and manual penetration testing. Automated vulnerability scanning (Qualys) runs weekly against all externally-accessible endpoints and monthly against internal infrastructure. Results are triaged by the security team with critical findings requiring remediation within 72 hours and high findings within 30 days.

Annual penetration testing is conducted by an independent third-party security firm with experience in financial services environments. The scope includes external network penetration, internal network penetration, application-level testing of the API gateway and web services, and social engineering exercises targeting operations staff. The most recent penetration test (November 2025) identified 2 medium-severity findings, both of which were remediated within 15 business days.

16.4 Compliance Validation

Compliance validation is an ongoing process rather than a point-in-time assessment. Continuous compliance monitoring tools validate that infrastructure configurations remain aligned with PCI-DSS, SOX, and OCC requirements. Drift detection alerts fire when any configuration change moves a system outside its approved security baseline, enabling

rapid remediation before the drift creates a compliance gap.

The annual PCI-DSS assessment by the Qualified Security Assessor (QSA) is prepared through a combination of automated evidence collection, quarterly internal scans, and pre-assessment readiness reviews conducted 60 days before the formal assessment. This preparation process has resulted in zero failed assessments since the platform's initial certification in 2019.

17 References

- IBM z/OS Parallel Sysplex documentation